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Charge Transfer-Induced SERS Enhancement of MoS_2/Dopants Dependent on their Interaction Difference

Lei Chen, Juan Pedro Merino, Miquel Torrent-Sucarrat, Hui-Lei Hou, and Maurizio Prato**

2D transition metal dichalcogenide materials have attracted increasing attention as active surface-enhanced Raman spectroscopy (SERS) platforms. In this study, the influence of n- and p-type doping of exfoliated MoS_2 ($exMoS_2$) hybrids on the SERS performance is investigated, employing Rhodamine 6G (R6G) as a probe molecule. It is demonstrated that n-doped $exMoS_2$ hybrids ($exMoS_2$ mixed with C_{60}, graphene, and sodium dodecyl sulfate) exhibit enhanced SERS intensities, while p-doping ($exMoS_2$ mixed with TCNQ) resulted in inhibited SERS enhancement. A key discovery is the linear relationship between Raman enhancement of MoS_2/dopant hybrids and the difference in their LUMO energy levels, which dictate the degree and direction of charge transfer. Interestingly, MC_{60}-4, a C_{60}-doped hybrid, deviates from the linear relationship, displaying remarkable SERS enhancement owing to its chemical interaction and unique Raman scattering activity. The findings provide critical insights into the SERS enhancement behavior of doped MoS_2, facilitating precise tuning of SERS intensities by manipulating the MoS_2 doping state.

1. Introduction

Surface-enhanced Raman spectroscopy (SERS) has attracted tremendous attention due to its high sensitivity and non-destructive analysis for detecting trace species and single molecules.[1–5] Although the precise mechanism of the enhanced effect is still under argument, conventional SERS enhancement includes mainly two mechanisms, I) the electromagnetic mechanism (EM) generated by local surface plasmon resonance (LSPR) upon laser excitation at the surface of a noble metal nanostructure,[6] and ii) the chemical mechanism (CM), which arises from the chemical interaction and charge transfer (CT) between the analyte and the substrate.[7–9] Considering the above mechanisms, the SERS enhancement effect (EF) strongly depends on the substrate.[10] Therefore, SERS substrates with high EFs have been intensively studied in the past few decades.[3,11–13] Generally, analytes are adsorbed on rough metal surfaces, dominating SERS experiments. Based on EM, noble metals (Au and Ag) and their hybrids are well studied with high SERS EFs[14–17] due to their plasmonic resonance frequencies matching the excitation wavelength range commonly used for Raman spectroscopy. Unfortunately, the high cost and low stability of noble metals limit the practical applications of SERS. Thus, searching for alternative materials as SERS substrates is urgently requested.

Since graphene was discovered in 2004,[18] 2D materials (2DMs) have been widely studied in various fields, such as electrocatalysis,[19] gas sensors,[20] and pressure sensors.[20] Recently, 2DMs, for example, graphene[21–24] and 2D transition metal dichalcogenides (TMDs),[25–27] have been developed as SERS substrates other than noble metals based on CM. MoS_2 is one of the most intensively studied materials among

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TMDs due to its unique physical and chemical properties for SERS-mediated molecule detection.[28–30] The 2H phase MoS_2 (2H-MoS_2) is a polytype with semiconducting properties.[31] Moreover, the energy band levels of 2H-MoS_2 can be tuned by various doping, such as plasma bombardment,[32,33] atom substitution,[34] surface charge transfer,[35–37] which plays a critical role in semiconductor-based SERS substrates.[38] Surface charge transfers are interesting because the dopants are not in the transport pathways of electrical carriers, and the van der Waals (vdW) interactions between 2DM and dopants generally do not damage the 2DM lattice.[39] The charge transfer direction is from a higher lowest unoccupied molecular orbital (LUMO) or fermi level material to a lower one.[40] Therefore, organic or inorganic substances can be used as n or p dopants to modulate the electronic property of MoS_2. For example, amino acids can cause reversible n-type surface functionalization of MoS_2 monolayer,[41] and the bandgap of MoS_2 decreases with the adsorption of amino acids, with the higher highest occupied molecular orbital (HOMO) of amino acids and the smaller bandgap of amino acid/MoS_2.[42] Tetracyanoquinodimethane (TCNQ),[43] tetracyanoethylene (TCNE),[43] and O_2[44] are p dopants for MoS_2. In addition, Yang et al., based on density functional theory, have reported that charge transfer between 2D materials and dopants linearly depends on the HOMO of n-type dopants and 2D conduction band minimum state or the LUMO of p-type dopants and 2D valence band maximum state.[45] Nonetheless, an effective and low-cost approach for accurately modulating the carrier type and density in semiconducting 2DMs is still challenging in constructing practical 2DM devices.[32,46,47] Furthermore, the impact of doping-induced modulation on the energy band structures of 2H-MoS_2 on SERS remains uncertain.

Herein, we systematically study the Raman enhancement effect of MoS_2 based on the CT between the dopants and MoS_2 with Rhodamine 6G (R6G) as a probe molecule. An efficient, facile, and low-cost approach was developed for tuning energy band structures of 2H-MoS_2 in MoS_2/dopants hybrids, which are obtained by a physical mixture of exfoliated MoS_2 with TCNQ, graphene (Gr), Sodium dodecyl sulfate (SDS), and C_{60}, respectively. The obtained MoS_2/dopants hybrids display Raman enhancement depending on CT dynamics: where dopants transfer electrons to MoS_2 enhancing the Raman signal, while dopants extracting electrons from MoS_2 inhibit the Raman enhancement effect. The identification of a linear correlation between the Raman enhancement observed in MoS_2/dopant hybrids and the disparity in their lowest unoccupied molecular orbital (LUMO) energy levels, except the MC_{60}-4 diverges from this linear trend. Density Functional Theory (DFT) calculations were then conducted to analyze the stabilization energies, charge distributions, and Raman and infrared intensities of dopant-R6G complexes. The computational Raman spectra of the dopant···R6G complexes confirm that the C_{60}···R6G interaction exhibits notably higher derivatives of polarizability. SDS···R6G triggers the most significant charge transfer distribution from the dopant to the probe molecule, resulting in MoS_2/SDS hybrid displays SERS EF as high as 4.4×10^4, comparable to the 1T phase MoS_2. This work provides an in-depth understanding of the interplay between SERS and the doping status of $exMoS_2$, facilitating accurate regulation and prediction of SERS intensity. Additionally, it paves the way for novel approaches to enhance SERS performance by strategically altering the organic dopant structure and incorporating additional polarizable substituents.

2. Results and Discussion

Exfoliated MoS_2 ($exMoS_2$) was obtained by sonication of bulk MoS_2 in NMP followed by centrifugation, as detailed in the Supporting information. Statistical analysis of 150 flakes revealed that ≈85% of MoS_2 flakes displayed lateral dimensions ranging from 50 and 200 nm (Figure S1, Supporting Information). **Figure 1a** displays a typical tapping mode atomic force microscopy (AFM) image of an exfoliated MoS_2 drop-cast on a SiO_2/Si substrate. Additionally, Figures 1b,c illustrate that ≈70% of $exMoS_2$ sheets with a similar topographic AFM height from 1.6 to 2.0 nm indicating a bilayer structure, although a few thinner or thicker nanosheets (≈0.8 nm in height for monolayer, 2.3 nm for a trilayer)[10] can also be found. Transmission electron microscopy (TEM) further confirmed the successful exfoliation of MoS_2 (Figure S2, Supporting Information). Furthermore, the bilayer structure of MoS_2 suspended on a lacey carbon film was further confirmed by high-resolution TEM (HRTEM) at the edge, with a parallel fringe spacing of 0.62 nm (Figure 1d). The lattice structure remains intact after the ultrasonic exfoliation, and the lattice d spacing is estimated to be 0.27 nm, corresponding to the (100) lattice plane of the 2H MoS_2 phase.[48] Specifically, $exMoS_2$ displayed the characteristic Raman peaks of the 2H phase (Figure S3, Supporting Information) at ≈380, ≈405, and ≈450 cm^{-1}, arising from the E^1_{2g} (in-plane), A_{1g} (out-of-plane) vibration modes of MoS_2, and the typical second-order longitudinal acoustic phonon peak (2LA(M)), respectively. The presence of excitonic A and B bands in the UV-Vis spectrum of $exMoS_2$ indicates it maintains the 2H phase,[49] as shown in Figure S4 (Supporting Information).

Fullerene (C_{60}) was first chosen as the electron donor for doping MoS_2 to generate n-doped MoS_2 because the higher LUMO of C_{60} (−3.8 eV) than that of MoS_2 (−4.5 eV) can result in a charge transfer from C_{60} to MoS_2 in C_{60}/MoS_2 (MC_{60}) hybrids.[50] In practice, under laser irradiation, the HOMO electrons can be excited to the LUMO level, and then the higher LUMO materials (excited state) can transfer electrons to the lower LUMO materials. The Raman spectra of C_{60}, $exMoS_2$, and MC_{60} hybrids with different amounts of additive C_{60} obtained by drop-coating on SiO_2/Si wafer are displayed in **Figure 2a**. All the MC_{60} hybrids display blue shifts for both E^1_{2g} and A_{1g} peaks due to strain or doping-induced changes in the electronic structure.[51] The characteristic peak of C_{60} is not evident in all MC_{60} hybrids due to its small amount and charge transfer between MoS_2 and C_{60}.[52] Scanning electron microscopy (SEM) was performed to characterize the morphologies of C_{60}, $exMoS_2$, and MC_{60} hybrids. The layer structure of pure $exMoS_2$ and irregular block C_{60} were observed (Figure S5, Supporting Information). Figures 2b and Figure S5 (Supporting Information) display similar morphologies for all MC_{60} hybrids, losing the clear layer structure compared to $exMoS_2$. The morphology change indicates the strong interaction between C_{60} and $exMoS_2$.

The Raman peak located at 613 cm^{-1} was utilized to evaluate the intensity of the SERS signal. Figure 2c shows the SERS intensities of each sample after incubating 2 h with R6G as the probe molecule. The SERS intensities of MC_{60} hybrids versus the

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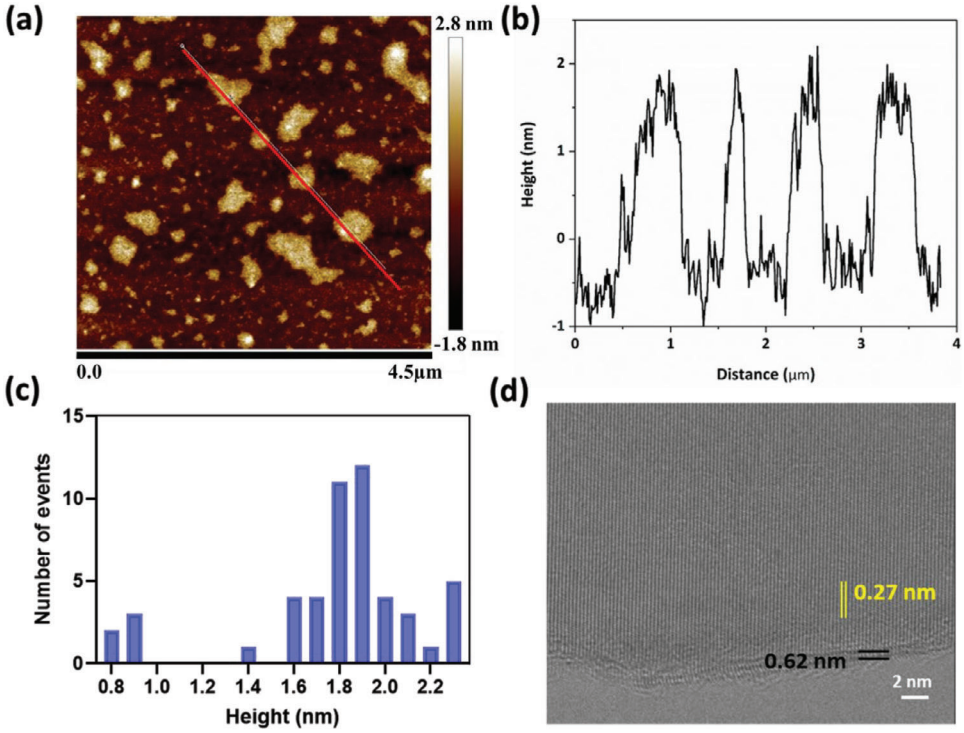


Figure 1. a) AFM image of the exMoS$_2$ on SiO$_2$/Si substrate, b) the corresponding line scan in (a). c) AFM height distribution of exMoS$_2$. d) HRTEM of the obtained at the edge of exMoS$_2$.

amount of additive C_{60} are plotted in Figure 2d (MoS_2 is 1 mg for all hybrids). Pure C_{60} displayed very low SERS intensity, and enhanced SERS intensities regarding pure exMoS$_2$ were observed with C_{60} doping (Figure 2c). Interestingly, the SERS intensity increases with the amount of added C_{60} (from MC_{60}-1 to MC_{60}-4) up to the mass ratio of 1:0.45 (MoS_2:C_{60}). Subsequently, the SERS intensity decreased with increased C_{60} content (MC_{60}-5). In comparison to exMoS$_2$, MC_{60}-5 demonstrated comparable intensity but displayed enhanced uniformity. This phenomenon can be explained as the charge transfer is saturated from C_{60} to MoS_2 when C_{60} covers all the MoS_2 surface (MC_{60}-4), and the excess C_{60} reduces the proportion of MoS_2, decreasing SERS intensity. Therefore, the highest SERS can be achieved by the optimal amount of dopant for MoS_2 hybrids when C_{60} precisely occupies the surface of MoS_2. The dispersion of values in each measurement is because the MoS_2 or MoS_2/dopants cannot form a uniform film during the drop-coating process, which is a direction that needs further study and improvement.

With this information in hand, we then used TCNQ, graphene (bilayer, Figure S6, Supporting Information), and SDS as dopants to MoS_2, noted as MTCNQ, MGr, and MSDS, respectively. The Raman spectra of MoS_2/dopants hybrids displayed blue shifts of E^1_{2g} and A_{1g} compared to exMoS$_2$ (**Figure 3a**), indicating the interaction between MoS_2 and dopants.[51] The results agree with MC_{60} hybrids due to doping or strain. The morphologies of these hybrids were characterized by SEM. MGr hybrid maintained a layer structure similar to exfoliated graphene and MoS_2 (Figure S7, Supporting Information). However, the morphologies MC_{60}-4, MTCNQ, and MSDS hybrids are more like particles rather than layer structures (Figures 2b, 3b; Figure S7, Supporting Information). The results reveal that doping MoS_2 with different dopants can change its morphology due to their strong interaction. In Figure 3c, the SERS intensity with R6G as the probe molecule increases for MGr, MC_{60}-4, and MSDS while decreasing for MTCNQ, demonstrating a different effect of TCNQ on MoS_2 Raman enhancement from the other dopants. The 2D diagram of C_{60}, MC_{60} hybrids, MGr, exMoS$_2$, MTCNQ, and MSDS average Raman spectra are provided in Figures S8 and S9 (Supporting Information) to facilitate clearer comparison. The LUMOs of these five materials are listed in **Table 1**. The LUMO of MoS_2 (−4.5 eV)[50] is lower than those of Gr, C_{60}, and SDS (see Table 1) but higher than that of TCNQ (−5.1 eV). Therefore, CT from dopants to MoS_2 in MGr, MC_{60}-4, and MSDS hybrids results in n-type doping of MoS_2, while the opposite direction of CT in MTCNQ leads to p-type doping of MoS_2. The results indicate that n-doping of MoS_2 enhances, while p-doping inhibits SERS of MoS_2.

Additionally, the potential difference between MoS_2 and dopants further confirms the doping-induced CT that enhances or inhibits SERS. The flat band potential of MoS_2 (− 0.13 eV vs Standard Hydrogen Electrode (SHE))[53] is lower than the reduction potential of TCNQ (0.46 V vs SHE)[54] and higher than C_{60} (−0.43 V vs SHE).[55] This demonstrates that MoS_2 can be oxidized by TCNQ but reduced by C_{60} because the charge transfer is from a low potential to a higher one. Therefore, TCNQ, as a p-type dopant, extracts electrons from MoS_2, while C_{60} donates electrons to MoS_2, consistent with the results of LUMO.

In this context, the Raman enhancement can be explained by the difference in the LUMO energy levels between MoS_2 and dopants (represented as ΔLUMO),[45] which determines the

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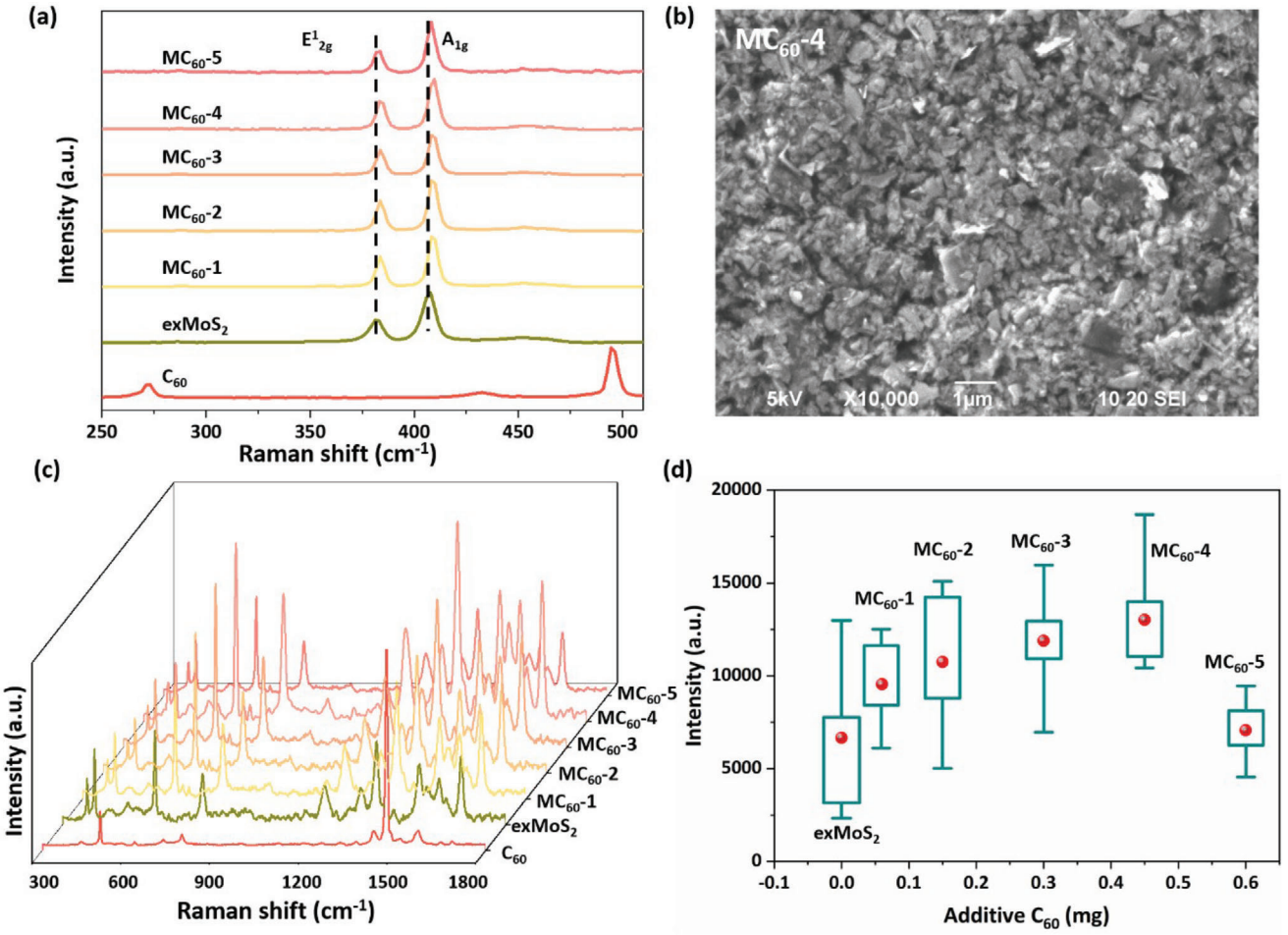


Figure 2. a) Raman spectra of C_{60}, MoS_2, and MC_{60} hybrids with a laser of 532 nm. b) SEM of MC_{60}-4. c) Average Raman spectra of C_{60}, MoS_2, and MC_{60} hybrids with R6G (10^{-4} M) as the probe molecule under 532 nm laser. d) Plotting of SERS intensity of $exMoS_2$ and MC_{60} hybrids versus the amount of additive C_{60}. The red points are the average intensity of each sample. MoS_2 is 1 mg for all hybrids.

degree and direction of charge transfer in MoS_2/dopant hybrids, and is defined as

$$\Delta\text{LUMO} = \text{LUMO (dopant)} - \text{LUMO}\left(MoS_2\right) \quad (1)$$

A positive value of ΔLUMO indicates CT from the dopant to $exMoS_2$, while a negative value signifies charge transfer in the opposite direction. Figure 3d displays a modest correlation between SERS intensity and ΔLUMO. However, excluding the MC_{60}-4 outlier from the analysis reveals a strong correlation, with an R^2 value of 0.99. Raman spectroscopy is sensitive to changes in the material polarizability, which can be influenced by its electronic structure.[56] The difference in the LUMO energy levels between pure MoS_2 and the dopants indicates how the dopant modifies the electronic structure of the host material, which alters polarizability and affects Raman intensities. The high linear correlation between ΔLUMO and experimental Raman intensities in doped MoS_2 can be attributed to the dopant-induced changes in the electronic structure.

In order to explain this phenomenon, we focused on investigating the CT mechanism between the dopants and R6G, since it may provide valuable insights into understanding the remarkable SERS enhancement effect exhibited by MC_{60}-4. Using DFT calculations (for more details, see Supporting Information), four complexes formed between the four dopants (graphene has been modelized as $C_{54}H_{18}$ with D_{6h} symmetry) and R6G have been studied (see Figure S10, Supporting Information). The free energy stabilization of these SDS···R6G, Gr···R6G, C_{60}···R6G, and TCNQ···R6G complexes are −67.0, −26.7, −7.3, and −5.7 kcal mol^{-1}, respectively (see Table S1, Supporting Information).

The combination of electrostatic interactions between a negative anion (sodium dodecyl sulfate) and a positively charged structure (R6G), along with the presence of a short hydrogen bond (1.91 Å between (SDS)S-O^-···H-N(R6G), as well as several C-H···π interactions, culminates in the formation of the most stable complex, SDS···R6G (refer to Figures S10 and S11, Supporting Information). This structure also triggers the most significant charge transfer distribution from the dopant to the probe molecule, providing a rationale for the pronounced Raman EF observed for MSDS. The other dopants engage in comparable non-covalent interactions, specifically π–π stacking.

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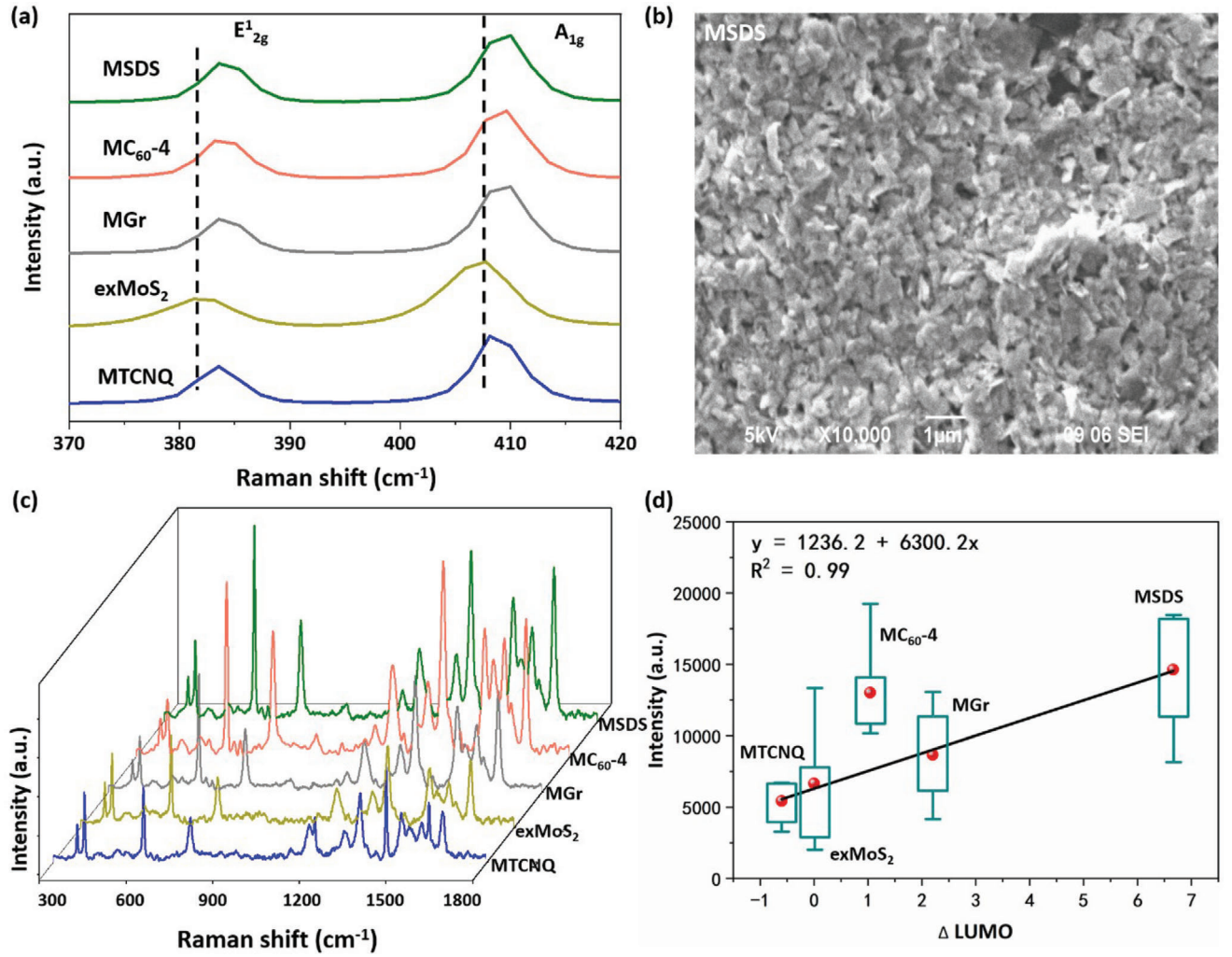


Figure 3. a) Raman spectra of MTCNQ, $exMoS_2$, MGr, MC_{60}-4, and MSDS hybrids with a laser of 532 nm. b) SEM of MSDS. c) SERS intensity of MTCNQ, $exMoS_2$, MGr, MC_{60}-4, and MSDS hybrids with R6G (10^{-4} M) as the probe molecule under 532 nm laser. d) Plotting of SERS intensity of each hybrid versus the ΔLUMO. The red points are the average intensity of each sample.

In the graphene case, the aromatic rings of R6G align parallel with the graphene dopant, establishing robust π-π interactions at distances less than 3.5 Å.[57] This results in a binding free energy surpassing 25 kcal mol^{-1}. Conversely, in the TCNQ···R6G complex, less effective non-covalent interactions form, consistent with the antiaromatic character of TCNQ, leading to the weakest free energy stabilization. Finally, the curvature of C_{60} diminishes its effective surface area for π-π stacking, yet it maintains relatively short distances of ≈ 3.2 Å i relation to R6G. Furthermore, it disrupts the planarity of the R6G multi-aromatic rings. Overall, this conformation results in a viable binding free energy of ≈ 7 kcal mol^{-1}. These observations align with the pattern that the strongest binding stabilization between the dopant and R6G corresponds to the highest SERS enhancement, except for the C_{60} dopant.

Furthermore, scrutinizing the charge distribution within these complexes reveals that SDS, Gr, and C_{60} dopants redistribute electron density to the R6G unit. In contrast, the TCNQ molecule can withdraw electron density from the R6G unit, as outlined in Table S1 (Supporting Information). These findings are in line with experimental observations of SERS enhancements and reductions achieved using n- and p-dopants, respectively. It is noteworthy that among the n-dopants, C_{60} exhibits the least electron-donating characteristics. Consequently, the CT mechanism between the dopants and R6G inadequately explains the remarkable SERS enhancement factor displayed by MC_{60}-4, necessitating an alternative rationalization.

Table 1. LUMO[a] and Potential of TCNQ, $exMoS_2$, Gr, C_{60}, and SDS.

	TCNQ	$exMoS_2$	Gr	C_{60}	SDS
LUMO (eV)	−5.10	−4.50[42]	−2.30	−3.46	2.16
Potential (V vs SHE)	0.46[45]	−0.13[44]	–[b]	−0.43[46]	–[b]

[a] If it is not indicated otherwise, evaluated at B3LYP-D3BJ/6-311G(2d,2p) level of theory (for additional details see Supporting Information). [b] Not found in the literature.

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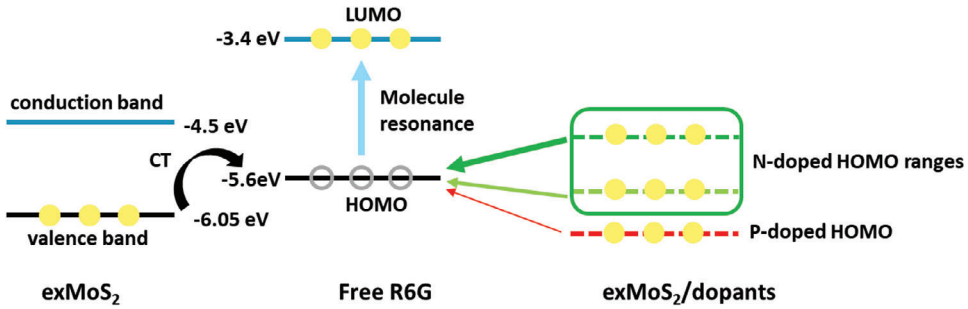


Figure 4. Schematic illustration of the energy band diagrams and charge transfer process from $exMoS_2$ and $exMoS_2$/dopants to R6G. The HOMO and LUMO levels of free R6G, the HOMO (valence bands), and the LUMO (conduction bands) of $exMoS_2$ and $exMoS_2$/dopants are shown in the plot. The yellow dots and grey circles represent the electrons and holes, respectively. Arrows represent molecule resonance and charge transfer, respectively. The dashed line represents the shifted HOMOs due to doping.

Raman scattering requires a change in the polarizability for vibrational excitation to occur, that is, the polarizability derivative to the normal coordinate associated with the vibration must be nonzero. Building on this premise, we postulate that the C_{60}···R6G interaction yields larger electric dipole-electric dipole polarizability derivatives compared to the other complexes. The SERS intensities illustrated in Figures 2d and 3d correspond to the vibrational coordinate linked with the 613 cm^{-1} mode, which pertains to the C−C ring bending mode (in-plane) within the phenyl rings of the probe. This specific mode was chosen due to its high sensitivity to even minor alternations in the chemical interaction between R6G and the MoS_2/dopant surface. Furthermore, this hypothesis finds validation through DFT calculations (refer to Figure S13, Supporting Information). Regarding the studied vibrational mode, the Raman scattering activity of the C_{60}···R6G complex exhibits the most substantial value, 80 $Å^4$ amu^{-1}. For the sake of comparison, the Gr···R6G complex registers a value of 12 $Å^4$ amu^{-1}.

Reflecting upon the results outlined above (and considering MC_{60}-4 as an isolated case), the possible CT mechanisms for Raman enhancement of MoS_2/dopants hybrids are proposed better to understand the doping effect (**Figure 4**). The Raman signal of the bulk R6G is attributed to the molecule resonance induced by the electronic transition from the HOMO (−5.70 eV) to the LUMO (−3.40 eV)[48] due to its bandgap fitting well with the laser energy (2.33 eV). When MoS_2 or MoS_2/dopants hybrids were used as the SERS substrate, the electron transfer mechanism can be delineated through a two-step process. In the initial step (Process I), the electron excitation occurs from the HOMO energy level to the LUMO energy level of R6G, thereby generating holes in the HOMO. Subsequently, CT happened from the HOMO of MoS_2 to that of R6G (process II) leading to recombining with holes induced by excitation of molecule resonance for a significant Raman signal.[10] Moreover, the lower additional energy required for process II results in a stronger Raman EF. For n-type dopants, two situations are considered. In the first case, the N-doped HOMO is higher than that of R6G. The n-type dopants' electrons are dropped down directly to the HOMO level of R6G without requiring extra energy in process II. This direct transition enables efficient charge transfer and yields significant enhancement in SERS signals. In the second situation, the N-doped HOMO is slightly lower than that of R6G, which leads to a lower energy gap between the $exMoS_2$/dopants and R6G and improves the CT from MoS_2 to R6G.[58] On the contrary, p-type dopants extract electrons from MoS_2, decreasing the HOMO level of MoS_2 and increasing the extra energy required for CT between MoS_2 and R6G, which inhibits CT from MoS_2 to R6G and reduces SERS enhancement. In other words, n-doped MoS_2 with higher HOMO requires smaller extra energy or without extra energy for the CT from MoS_2 to R6G, leading to stronger Raman enhancement, and p-doped MoS_2 with lower HOMO requires more extra energy resulting in weak Raman enhancement.

Figure 5a depicts the Raman signals obtained from MSDS hybrid samples with varying concentrations of R6G to determine the limit of detection. Compared to the MSDS with 0 M R6G (Figure 5b), the Raman signal at 613 cm^{-1} remains detectable even at a low concentration of 10^{-7} M (Figure 5c). The limit of detection is 1×10^{-7} M with the Raman enhanced factor of 4.4×10^4, comparable to 1T MoS_2 (2.9×10^4).[26] Additionally, the EF values of MTCNQ, $exMoS_2$, MGr, and MC_{60}-4 are 1.6×10^4, 1.9×10^4, 2.5×10^4, and 3.8×10^4, respectively.

3. Conclusion

In summary, we have demonstrated that n- and p-doped MoS_2 can improve and inhibit Raman enhancement via CT mechanism with R6G as the probe molecule, respectively. Because the HOMO of MoS_2 can shift up to reduce the extra energy required for CT from MoS_2 to Rhodamine 6G in n-doped hybrids (MC_{60}, MGr, MSDS), while the opposite is for p-doped hybrid (MTCNQ). In addition, the MSDS hybrid displayed SERS EF as high as 4.4×10^4. Moreover, the Raman enhancement of MoS_2/dopants hybrids linearly depends on their LUMO difference, which determines the degree and direction of CT. Notably, MC_{60}-4 emerges as a conspicuous outlier within this CT mechanism, showcasing a significant SERS EF despite having a moderate LUMO difference between C_{60} and MoS_2. The computational Raman spectra of the dopant···R6G complexes validate that the C_{60}···R6G interaction displays more substantial derivatives of polarizability concerning the vibrational mode used for the SERS measures, in contrast to the other complexes. This study offers comprehensive insights into the relationship between SERS and the doping state of MoS_2, enabling precise control and prediction of SERS intensity. Furthermore, it opens up new possibilities for enhancing SERS performance through strategic modifications of the organic dopant structure, such as C_{60}, and adding more polarizable substituents.

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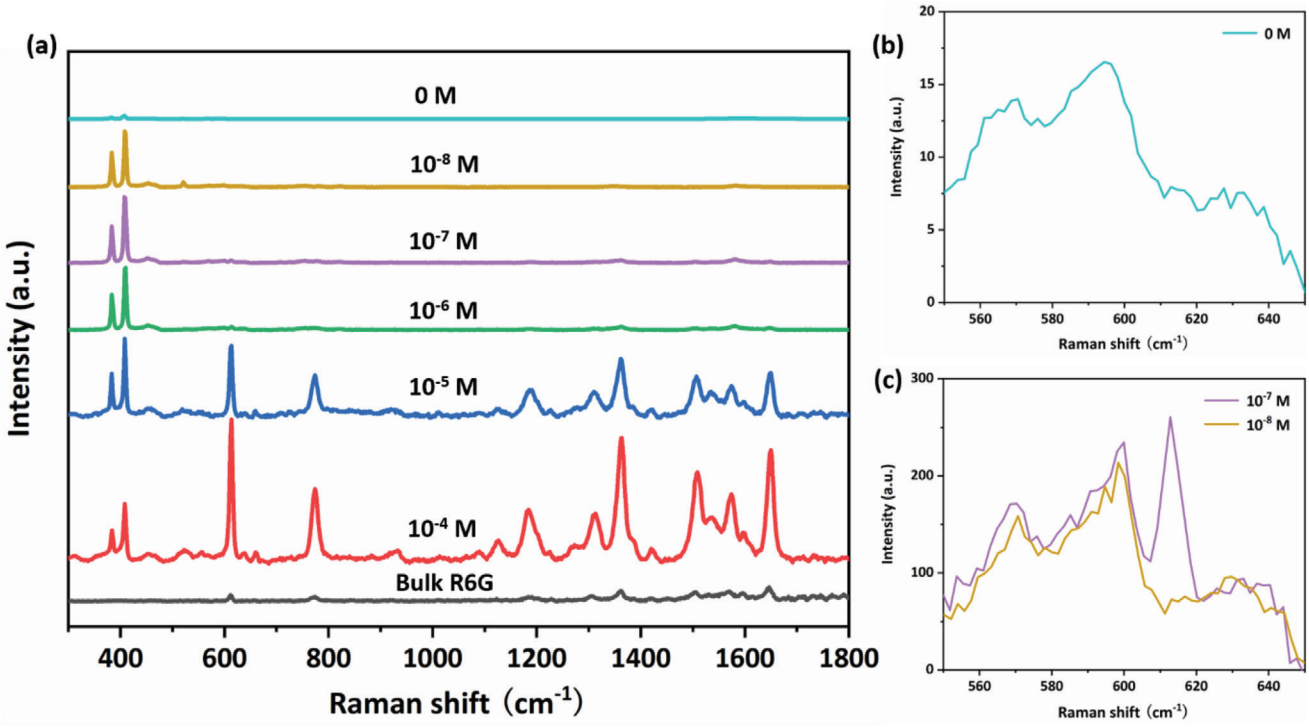


Figure 5. a) SERS measurements based on MSDS hybrid were performed with different concentrations of R6G with a 532 nm laser. b) Local enlarged image of MSDS with 0 M R6G. c) Local enlarged image of MSDS with 10^{-7} and 10^{-8} M R6G.

4. Experimental Section

Material and Reagents: Synthetic spherical graphite (SG17, ≥ 99.95% C, Bonding Chemical, USA) with tap density ≥1 g ml^{-1} and a specific surface area ≤ 6.5 m^2 g^{-1} was used after annealing under vacuum (300 °C). MoS_2 (≈6 µm) powder, tetracyanoquinodimethane (TCNQ), graphene (Gr), C_{60}, and Sodium dodecyl sulfate (SDS) were purchased from Sigma–Aldrich. All chemicals, including organic solvents used in the protocols, were purchased from either Sigma–Aldrich or Alfa-Aesar and used as received.

Characterizations: The exfoliated MoS_2 samples were characterized by various methods such as atomic force microscopy (AFM), Raman spectroscopy, UV–vis spectroscopy, transmission electron microscopy (TEM), and high-resolution transmission electron microscopy (HRTEM). Raman spectra were recorded with a Renishaw Invia Raman spectrometer equipped with a green laser (λ = 532 nm). UV–vis spectra were performed in a Jasco V-630 BIO spectrophotometer. TEM images were obtained using a JEOL JEM-2010 electron microscope at 120 kV equipped with a GATAN US1000 CCD camera. HRTEM images were obtained using a JEOL JEM-2100F UHR electron microscope at 200 kV equipped with a TVIPS F-216 CMOS camera. The TEM samples were prepared by drop-casting the dispersion of solutions onto a copper grid coated with an ultra-thin carbon film (Ted Pella Inc., USA) and dried in ambient conditions. Surface topologies were characterized by atomic force microscopy (AFM) in intermittent contact tapping mode using a tapping etched silicon probe (TESPA-V2) with a resonant frequency and spring constant of 320 kHz and 42 N m^{-1}, respectively. Samples were prepared by drop-casting of an exfoliated MoS_2 suspension on a SiO_2/Si substrate. The obtained AFM images were analyzed by NanoScope Analysis 1.9.

Glove Box: Sample synthesis and preparation was carried out in an argon-filled glovebox, equipped with a gas purifier and solvent vapor removal unit: oxygen content <0.1 ppm, water content <0.1 ppm.

Exfoliation of MoS_2: MoS_2 powder was dried at 150 °C under vacuum overnight prior to use. The exfoliated MoS_2 was prepared by liquid-phase sonication. Bulk MoS_2 (2 g) was put into 200 mL 1-Methyl-2-pyrrolidone (NMP) and sonicated for 1 h. Then, MoS_2 dispersion was centrifugated at 2000 rpm for 30 min. The resulting supernatant was collected, affording exMoS_2 (0.08 mg mL^{-1}).

Exfoliation of Graphene: Exfoliated graphene was prepared as previous report.[59] Pristine spherical graphite (SG17) was heated at 300 °C under vacuum for 3 days and transferred into the glovebox with Ar atmosphere. Then 72 mg (6 mmol C) graphite and 30 mg (0.75 mmol K) potassium were mixed and heated at 180 °C overnight to obtain the brown golden KC_8 graphite potassium intercalation compound. Then, the resulting KC_8 (102 mg) was dissolved in 100 mL of anhydrous THF at room temperature in a glovebox, then it was exported from the glovebox to 2 h sonication (water bath). Subsequently, graphene dispersion was centrifugated at 3000 rpm for 20 min. The resulting supernatant was collected in a glovebox, and was quenched the negative charge by the addition of 10 mL PhCN. Finally, 50 mL cyclohexane was added to the solution, which was then washed three times with 50 mL water. The organic phase was filtered over a reinforced cellulose membrane filter (0.45 µm pore). The grey solid was washed with acetone, THF, toluene, and DMF.

Preparation of MC_{60} Hybrids: The toluene solution (1 mL) of C_{60} (0.06, 0.15, 0.30, 0.45, and 0.60 mg) was put into exMoS_2 dispersion NMP (1 mL, 1 mg mL^{-1}) and sonicated 10 min to afford MC_{60}-1, MC_{60}-2, MC_{60}-3, MC_{60}-4, and MC_{60}-5.

Preparation of MGr Hybrid: 0.25 mL exfoliated graphene NMP dispersion (1 mg mL^{-1}) was put into exMoS_2 dispersion NMP (1 mL, 1 mg mL^{-1}) and sonicated 10 min to afford MGr.

Preparation of MTCNQ Hybrid: 9 mL acetone containing 0.32 mg of TCNQ was put into exMoS_2 dispersion NMP (1 mL, 1 mg mL^{-1}) and sonicated 10 min to afford MTCNQ.

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Preparation of MSDS Hybrid: 9 mL acetone containing 0.41 mg of SDS was put into exMoS$_2$ dispersion NMP (1 mL, 1 mg mL^{-1}) and sonicated 10 min to afford MSDS.

Preparation of SERS Substrates with Hybrids: The resulting dispersion obtained in the above procedures was drop-casted on SiO_2 wafers and dried at ≈ 70 °C.

SERS Measurements: The obtained SERS substrates were incubated in the different concentrations of R6G solutions for 2 h. The R6G deposited exMoS$_2$ or hybrids substrates were washed with distilled water to remove the free R6G and dried under an air atmosphere. The SERS spectra of so-prepared substrates were recorded using a 100× objective (NA = 0.85) at excitation wavelengths of 532 nm (0.13 mW power laser on the surfaces, 30 s of integration time). The presented spectra were baseline-corrected and normalized to the same laser power and integration time. Except for the bulk R6G, its integration time was 3 s to avoid the overflow of the detection. For exMoS$_2$, MGr, MC$_{60}$ hybrids, MTCNQ, all the SERS spectra were obtained with 10^{-4} M of R6G. For MSDS, R6G concentrations are in the range from 10^{-4} to 10^{-7} M. The enhancement factor (EF) value of the fabricated MSDS substrates was estimated according to Equation S5 (Supporting Information).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

charge transfer, doping, molybdenum disulfide (MoS$_2$), Rhodamine 6G (R6G), Surface-enhanced Raman spectroscopy (SERS)

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